Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 43

Algorithms and Time Complexity

Covers Chartrand & Zhang, Sections 6.1 and 6.3

Status of this set

Ungraded practice. Solutions released with it. Show the count, not just the answer.

Part I: How to Practice This Material

Counting loop work

- A loop running 1 to n with one comparison inside: n comparisons.
- Independent nested loops: **multiply**.
- Sequential loops: **add**.
- An inner loop whose bound depends on the outer index: **sum** the inner counts.

That last case is where the Week 11 summation formulas earn their keep.

Three mistakes to expect

1. **Off by one on loop bounds.** "$i := 2$ to n" runs $n - 1$ times, not n.
2. **Multiplying when the inner bound varies.** That calls for a sum.
3. **Confusing worst case with average case.** Say which you are computing.

Part II: Practice Problems

Section A — Counting

1. Give the exact number of comparisons each structure performs on an input of size n.

(a) A single loop $i := 1$ to n with one comparison inside.

(b) Nested loops, outer $i := 1$ to n, inner $j := 1$ to n, one comparison inside.

(c) Outer $i := 1$ to n, inner $j := 1$ to i, one comparison inside.

(d) Outer $i := 1$ to $n - 1$, inner $j := i + 1$ to n, one comparison inside.

2. An algorithm finds both the largest and smallest number in a list of n numbers by comparing each new term first against the current maximum, and if that fails, against the current minimum.

(a) How many comparisons in the worst case?

(b) How many in the best case?

(c) Describe the input that produces each.

Section B — Worst, best, average

3. A linear search runs on a list of n distinct numbers.

(a) Best, worst, and average case when the target is present, each position equally likely.

(b) What is the worst case when the target is *absent*?

(c) For $n = 100$, give all four numbers.

Section C — Writing

4. In four to six complete sentences, explain why counting comparisons is a better measure of an algorithm than timing it on a particular computer.

Section D — Challenge (optional)

5. Show that finding both the maximum and the minimum can be done in about $\frac{3n}{2}$ comparisons rather than $2(n-1)$.